OLIVER WESTIN

ICT student at KTH · builds and runs self-hosted software

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Second-year **Information and Communications Technology** student at KTH who learns by building. I design, ship and run my own software end to end: a **media server that friends and family use every day**, low-level **Linux and Vulkan tooling** for gaming, and the tools behind a three-machine home network. At home anywhere from C and Rust to Node.js web apps and Windows server operations.

EXPERIENCE

Private Tutor · Allakando — Apr 2025 – Jul 2025

- Tutored upper-secondary students one-on-one, in person and online, in the subjects they found hardest, with personal study plans, homework and steady motivation.

Survey Interviewer · City of Stockholm — Jul 2024

- Interviewed residents and businesses across inner-city Stockholm about their finances in a three-week team survey; the data fed the city's decisions on improvements.

SELECTED PROJECTS

Marquee — 2026 – now

Self-hosted, phone-first media server that I run around the clock for friends and family: one-press downloads, Watch Together rooms, casting to TVs (DLNA/AirPlay), automatic MP4 conversion, subtitle auto-sync and frame-exact clip editing. Hardened after a security audit (sign-in, CSRF, least-privilege service); public release v1.1.

`Node.js 22 (no dependencies) · FFmpeg · SSE · qBittorrent API · TMDB · Windows service`

ember-latency — 2026

Vulkan layer that measures real click-to-photon latency in any game, from the input event to the frame on screen, live in MangoHud with no measurable FPS cost.

`C · Vulkan · VK_EXT_present_timing · Linux evdev`

Archfleet — 2026

Terminal dashboard for every machine at home over plain SSH: status, wake-on-LAN, processes, files, and one clipboard history shared between Linux and Windows.

`Python · Textual · SSH · PowerShell`

Clipper — 2026

Lightweight replacement for Medal/ShadowPlay: always-on replay buffer, review queue, trimmer, and clips stored on my own server instead of a cloud.

`Python · gpu-screen-recorder · OBS · FFmpeg`

agent-mesh — 2026

Gives AI coding agents (Claude Code, Codex) one shared set of instructions, memory and settings across three machines, synced through a git repo over SSH.

`Python · Git · SSH · systemd`

EDUCATION

KTH Royal Institute of Technology — Aug 2025 – Jun 2028

BSc, Information and Communications Technology (180 credits)

Stockholm Science & Innovation School — Aug 2022 – Jun 2025

Technology Programme, Information and Media Technology

Including Mathematics 5, Physics 2, Programming 1, Web Development 1, Computer and Network Technology, English 7.

SKILLS

Languages	Python, JavaScript / Node.js, C, Rust, Java, C#, PHP, Lua, SQL, Bash, PowerShell
Web	Next.js, React, HTML/CSS, REST and server-sent events, WebGL
Systems	Linux (Arch, Debian), systemd, Windows 11 as a server, networking, VPN, DNS, Secure Boot, Git, GitHub Actions
Media	FFmpeg, video encoding (H.264, HEVC, AV1), PipeWire audio
Spoken	Swedish (native), English (fluent)